

TECHNICAL DOCUMENT

Software and Research Tools

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Tools Actually Used

Tool	Version	Role	Official source
LTspice	26.0.2	Transistor-level simulation	analog.com/ltspice
Python	Bundled 3.x runtime	Verification, parsing, PDF and figure generation	python.org
Git	Installed client	Version control	git-scm.com
GitHub	Public repository service	Public dissemination	github.com
PTM	65-nm beta model	Predictive transistor model	mec.umn.edu/ptm
ReportLab	Bundled Python library	Academic PDF generation	reportlab.com
Pillow	Bundled Python library	Research figures	python-pillow.org

Reproducibility Notes

Versions and commands are documented in the repository. Verilog source is included, but no Verilog simulator result is claimed because Icarus Verilog was not installed during this run.